MATHEMATICS II **DATE:18/06/2025** **Period:8:30-11:30**

END OF TERM III EXAMINATIONS

GRADE SENIOR SIX

COMBINATIONS	►MATHEMATICS-COMPUTER SCIENCE-ECONOMICS (**MCE**) ►MATHEMATICS-ECONOMICS-GEOGRAPHY (**MEG**) ►PHYSICS-CHEMISTRY-MATHEMATICS **(PCM)** ►MATHEMATICS-CHEMISTRY-BIOLOGY **(MCB)** ►MATHEMATICS-PHYSICS-GEOGRAPHY **(MPG)** ►MATHEMATICS-PHYSICS-COMPUTER SCIENCE **(MPC)**

CANDIDATE NAMES: ....

INDEX NUMBER: ....

Instructions:

- the paper consists of two section **A** and **B:**

SECTIONA: Answer all questions from this section**(55marks)**

SECTION B: Answer only three questions from this section**(45marks)**

-Provide all answers on the appropriate answer sheet

-Complete in provide space and circle a letter corresponding the best answer

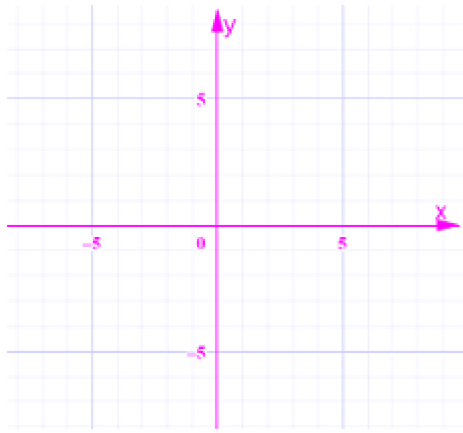
-The use scientific (non-programmable) calculators and geometric tools is allowed.

SECTION A: ATTEMPT ALL QUESTIONS (55marks)

1. Match column A to B **(5marks)**

Column A	Column B
A. The conics has a Centre, the semi major axis is always greater than the semi minor axis .	i. The parabola
B. It has an eccentricity greater than 1.	ii. Polar equation form of conics
C. It is a set of equidistant points from a fixed point and a fixed line.	iii. Canonical/standard equation form of conics
D. It displays the Centre, the semi major axis and the semi minor axis of conics.	iv. hyperbola
E. It has a focus at pole and the directrix at the focal parameter from the pole.	v. Ellipse

2. A committee of 5 is to be chosen from a list of 14 people, 6 of whom are men and 8 are women. their names are to be put in a hat and the 5 drawn out. What is the probability that this procedure a committee with (choose the correct answer)? **(3marks)**
   i. Exactly 3 men **a**. $\frac{36}{143}$ ***b***. $\frac{40}{143}$ ***c***. $\frac{43}{141}$ ***d***. $\frac{43}{140}$
   ii. At least two women ***a***. $\frac{134}{143}$ ***b***. $\frac{123}{143}$ ***c***. $\frac{94}{141}$ ***d***. $\frac{43}{141}$
   iii. More women than men ***a***. $\frac{49}{143}$ ***b***. $\frac{94}{143}$ ***c***. $\frac{57}{143}$ ***d***. $\frac{83}{143}$
3. The value of integral $\int_1^{e^2} lnx\, dx$ is: **(4marks)**
   a. $e^2 - 1$ **b**. $2e^2 - 1$ **c**. $2(e^2 - 1)$ **d**. 4 **e**. 3ln4
4. The conics has two Foci $(-5,2),(1,2)$ and eccentricity $\frac{2}{3}$ **(6marks)**
   i. This conic is **a**. Parabola **b**. Circle **c**. Hyperbola **d**. Ellipse
   ii. Determine its Centre, vertices and covertices
   Centre.........................vertices.......................covertices ....
   iii. Make the graph including Foci and directrices.



5. The value of y such that the vector $(1, y, 5)$ is a linear combination of vectors $\vec{u} = (1, -3, 2) and\ \vec{v} = (2, -1, 1)$ is: **(2marks)**

a. $y = 4$ **b.** $y = \frac{1}{6}$ **c.** $y = -8$

d. $y = 3$

6. The solution set of complex number equation $z^2 + (5 - i)z + 8 - i = 0$ is: **(4marks)**

a. $S = \{-2 + i, \quad 3 - 2i\}$ **b.** $S = \{-2 + i, \ -3 + 2i\}$

c. $S = \{-2 - i, -3 + 2i\}$

d. $S = \{3 + 2i, -2 + 2i\}$

7. The value(s) of k, such that the line $x + 2y = k$ is tangent to the ellipse $x^2 + 4y^2 = 8$ are **(4marks)**

a. $k = \pm 4$ **b.** $k = 6$ **c.** $k = \pm 2$ **d.** $k = \pm 3$

e. $k = \pm \frac{\sqrt{2}}{2}$

8. The loudness of a sound is measured as $L = 10 \log\left(\frac{I}{I_o}\right)$ decibels. Where I is the intensity and I_o is the reference intensity . if the intensity increases by a factor of 100 , how much does the loudness increase? **(3marks)**

a. $10\ dB$ **b.** $20\ dB$ ***c.*** $100\ dB$ ***d.*** $200\ dB$

9. If $X \sim B\ (10, 0.5). find\ \ P(X = 5)$ **(2marks)**

a. 0.246 ***b.*** 0.205 ***c.*** 0.312 ***d.*** 0.529

10. A discrete random variable X has the probability distribution shown in the table : **(6marks)**

X	6	7	8	9
$P(X = x)$	0.1	0.2	0.3	0.4

Choose the correct answer:

(i) $\Pr(X=8)=$ **a)** 0.6 **b)** 0.3 **c)** 0 **d)** None

(ii) $\Pr(X>7)=$ **a)** 0.7 **b)** 0.9 **c)** 0.2 **d)** None

(iii) $E(X)=$ **a)** 6 **b)** 7 **c)** 8 **d)** None

(iv) $Var(X)=$ **a)** 3.4 **b)** 2 **c)** 1 **d)** None

(v) $E(5-2X)=$ **a)** -10 **b)** -11 **c)** 11 **d)** None

(vi) $Var(5-2X)=$ **a)** 2 **b)** -3 **c)** 4 **d)** None

11. Find the length of the curve $y = \frac{e^x+e^{-x}}{2}$ between the points where $x = 0$ and $x = 2$ (choose the correct answer) **(3marks)**

a. $1 + 2e^2$ **b.** $\frac{e^2}{2} - \frac{1}{2e^2}$ c . 4

b. d. $\frac{1}{2}(e^x - e^{-x}) + c$

12. Consider the differential equation: $\frac{d^2y}{dx^2} - \frac{dy}{dx} + \frac{1}{4}y = 0$ admits the general solution:

Choose one from: **[4Marks]**

a) $y(x) = (C_1 + C_2 x)e^{2x}$

b) $y(x) = C_1 e^{-x} + C_2 e^{x}$

c) $y(x) = (C_1 + C_2 x)e^{\frac{1}{2}x}$

d) $y(x) = (C_1 \cos x + C_2 \sin x)e^{\frac{1}{2}x}$

13 . Choose the correct answer the integral of $\int_0^2 |2x-1|\,dx$ is **(3marks)**

a. $\frac{x^2}{2} - x + c$ **b.** $\frac{5}{2}$ **c.** $\ln(2x+1) + c$ **d.** $\frac{3}{2}$

14. If $\lim_{x\to-\infty}\left(\frac{a+\frac{1}{x}}{a-\frac{1}{x}}\right)^{-x} = e^{-1}$, then the value of a is **(2marks)**

a. -3 $\boldsymbol{b}$. 2 **c.** -2 **d.** $\frac{1}{2}$ **e.** none

15. Sugar dissolves in water at a rate proportional to the amount still undissolved. If there were $50kg$ of sugar present initially , and at the end of $5hours$ only $20kg$ of sugar is left, how much longer will it take until 90% of the sugar is dissolved. **(4marks)**

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SECTION B: ATTEMPT ONLY THREE QUESTIONS (45marks)

16. a) From a group of 4 men and 5 women , how many committees of assize 3 are possible.

i) With no restriction? **(3marks)**

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ii) With 1 man and 2 women ? **(3marks)**

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iii) With 2 men and 1 woman if at least a man must be in the committee. **(3marks)**

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b) If 3 books are picked at random from a shelf containing 5 novels, 3 books of poems, and a dictionary. What is the probability that:

i) The dictionary is picked **(3marks)**

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ii) 2 novels and 1 book of poems are selected **(3marks)**

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17. a) Given that $z = x + yi$, and $\left|\frac{z-1}{z+1}\right| = 2$ .Find the Cartesian equation of the locus of z and represent this locus by a sketch on the argand diagram. **(5marks)**

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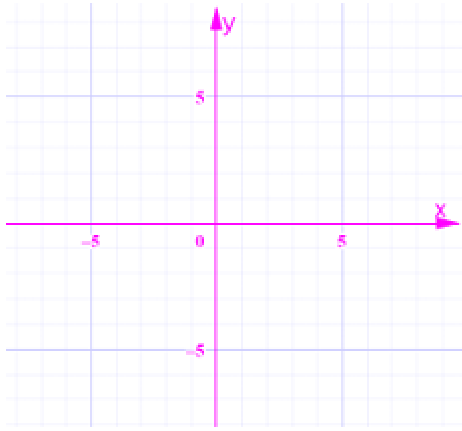
graph to represent locus



b) Given that $I_n = \int_0^{\frac{\pi}{2}} cos^n x \, dx, \;\; (n \geq 2)$

i) Find I_0 , I_1 and I_3 **(4marks)**

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ii) Write down the relationship between I_n and I_{n-2} **(4marks)**

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iii) Deduce I_4 and I_5 **(2marks)**

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18. Calculate

a) The surface area bounded by $X - axis$ and $\begin{cases} x = 6(\theta - sin\theta) \\ y = 6(1 - cos\theta) \end{cases}$, $0 \leq \theta \leq 2\pi$

(8marks)

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b) $\int sin^2x \, cos3x \, dx$ by using complex numbers. **(7marks)**

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19. Consider the curve $16x^2 - 9y^2 - 96x + 279 = 0$

a) Give the standard form of this conics and name it. **(3marks)**

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b) Find the Centre, the foci, the vertices and eccentricity **(6marks)**

centre……………………………

Foci………………………………

The vertices…………………………………

Eccentricity………………………………………

c) Find the length of the latus rectum **(1mark)**

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d) Determine the equation of asymptotes and directrices if they exist **(5marks)**

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20. Given statistics distribution

x	9	12	5	6	9	14	3	6	12	14
y	10	13	8	20	13	17	5	8	16	16

a) Complete the following table by appropriate values **(8marks)**

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$	R_x	R_y	d=$R_x - R_y$	d^2
$\sum x$ =	$\sum y$ =			$\sum(x-\bar{x})^2$ =	$\sum(y-\bar{y})^2$ =	$\sum(x-\bar{x})(y-\bar{y})$=				$\sum d^2$ =

b) Determine the variances, covariance's of two variables and correlation coefficient. **(3marks)**

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c) Determine the spearman's rank correlation coefficient of two variables. **(2marks)**

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d) Write down the regression line of the statistical distribution**(2marks**

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END OF EXAM